

ArtStation Breakdown Benchmark (F3)

1. Purpose of this module

This module aims to benchmark typical **breakdown** presentations on ArtStation or similar portfolios for technical art and real-time 3D projects. Because the environment does not allow live browsing, it uses general patterns and best practices observed in the industry rather than live examples.

The goal is to extract structural patterns and presentation techniques that Alexander can apply to his own **TwinSight X500** case study (and the Blender portrait) on ArtStation. These guidelines focus on: topology breakdown, material/shader breakdown, grooming breakdown, lighting breakdown, real-time asset optimisation, and technical process documentation.

2. Typical structure of a breakdown post

A polished breakdown often follows this structure:

Section	Content	Notes
Cover image	Eye-catching final render or hero image with project title	Sets context; simple caption (no clutter)
Introduction	Brief description (2–3 sentences) outlining the project goal, role, tools	Clarify personal contribution
Concept / references	Moodboard or concept art, if applicable	Show references and inspiration sources
Modelling / topology	Wireframe views, topology density, edge flow	Explain decisions (e.g. quads vs. tris, density)
UVs & baking	UV layout images, packing technique, notes on texel density	Mention tools (e.g. RizomUV, ZenUV)
Materials / shaders	Material breakdowns, shader graphs, texture maps	Include base colour, roughness, AO, normal maps
Lighting & rendering	Lighting setup diagrams, render passes	Explain environment or studio lighting choices
Real-time integration	Screenshots from Unity/Unreal; performance metrics	Show FPS, triangle counts, memory use
Tools and workflow	Summary of tools (Blender, Unity, Shader Graph, Substance)	Explain pipeline stages, CAD optimisation steps
Challenges & solutions	Brief mention of obstacles and how they were overcome	Keep honest; highlight problem-solving ability
Final thoughts	Lessons learned, future improvements	Avoid exaggerating scope

This structure can be adapted depending on the project. Not all sections are required, but they provide a comprehensive view of the work.

3. Observed patterns and takeaways

Even without live browsing, general trends in successful breakdowns include:

1. **Balanced visuals and text:** High-quality images take precedence, with concise captions or annotations. Long paragraphs are avoided; instead, bullet points or short sentences guide the viewer.
2. **Process transparency:** Artists show intermediate steps, such as blocked-out models, optimisation passes, wireframes, and shader graphs. This demonstrates technical depth and decision-making.
3. **Highlighting tools and pipeline:** Each tool used (Blender, Unity, Substance 3D Painter, RizomUV, etc.) is mentioned explicitly, often with logos or small icons.
4. **Performance metrics:** For real-time projects, metrics like triangle count, draw calls, frame rate, and memory usage are included to show optimisation awareness.
5. **Modular sections:** Each aspect (modelling, UVs, materials) is separated into its own subheading, making the post easy to navigate.
6. **GIFs and short videos:** Animated GIFs or embedded clips show effects (e.g. shader transitions, LOD swaps) without requiring external links.
7. **Consistent branding:** The creator uses a consistent color palette, typography, and layout across the breakdown. This reinforces professionalism.
8. **Role clarification:** Breakdowns specify individual contributions clearly, especially when multiple people worked on a project. This prevents misinterpretation.
9. **Reference to constraints:** For technical art, constraints (e.g. target platform, poly budget, performance targets) are stated. This contextualises design choices.
10. **Links to repositories or further reading:** Many posts provide GitHub links or documentation for those who want to delve deeper.

4. Suggestions for TwinSight X500 ArtStation breakdown

Based on the patterns above, Alexander should consider the following guidelines for his own breakdown:

1. **Clear cover and hook:** Use a high-quality hero image of the fully assembled drone or an exploded view. Include a short title and tagline (e.g., "TwinSight X500 – Real-Time Drone Assembly Visualisation").

2. **Concise introduction:** Outline the purpose (solving 2D documentation limitations), technologies used (Unity WebGL, Blender, C#, UI Toolkit), and role (sole developer / technical artist).
3. **Process segmentation:** Use subheadings for each phase: CAD import and evaluation, Blender cleanup and retopology, UV unwrapping, baking, shader and material setup, Unity optimisation, UI integration, performance metrics.
4. **Show wireframes and UV layouts:** Present before/after wireframes of CAD vs optimised models, and highlight how triangles were reduced from millions to ~95 k ¹. Show UV packing strategies.
5. **Shader graphs and visual modes:** Include snapshots of Shader Graph or code snippets illustrating how visual modes (X-Ray, blueprint, thermal style) were implemented.
6. **Performance and UX metrics:** Feature charts or infographics for key metrics (e.g., frame rate, memory usage). Reuse metrics from the thesis (SUS 91.88 and NASA-TLX results) to emphasise usability validation.
7. **Challenges & solutions:** Briefly mention obstacles (e.g., heavy CAD data, WebGL build size) and how they were addressed (instancing, LOD, baking, Scriptable Render Pipeline configuration).
8. **Future improvements:** Indicate potential enhancements (IoT integration, VR support, multi-asset support) without implying they already exist.
9. **External links:** Add a link to the GitHub repository, the portfolio page, and the demo video. Encourage viewers to try the WebGL build.
10. **Consistency with portfolio & GitHub:** Align fonts, colors, and layout across the portfolio site and GitHub README to unify the brand.

5. Relevance to Blender Portrait Breakdown

The same principles apply to the **Blender portrait** breakdown:

- Start with a strong render; provide concept art and reference images.
- Show sculpting stages (blocking, mid-level, detail), retopology, UVs, and material passes (albedo, roughness, subsurface scatter, displacement).
- Include grooming breakdown (hair/eyebrows setup) if applicable.
- Highlight how the portrait demonstrates technical art skills relevant to real-time engines (e.g., optimised topology, game-ready textures).

6. What to avoid

- **Overloading with text:** Avoid dense paragraphs; prefer bullet points and short sentences.
- **Skipping intermediate steps:** Don't jump directly from final render to finished asset; show the process.
- **Misrepresenting roles:** Don't claim full ownership of any parts created by others or by AI without clarifying assistance.
- **Exaggerated metrics:** Use only metrics that can be backed by the thesis or measured in the prototype; don't extrapolate to production contexts.

- **Broken links:** Ensure all external links work and lead to the correct page (GitHub, portfolio, video). Broken links reduce credibility.

7. Summary checklist

Before publishing an ArtStation breakdown, verify that:

- ☐ The cover image is polished and relevant.
- ☐ Each section has a clear subheading.
- ☐ Process images (wireframes, UV layouts, shader graphs) are included.
- ☐ Performance and usability metrics are shown with context.
- ☐ Tools and software used are credited.
- ☐ Personal contributions are clearly stated.
- ☐ External links (GitHub, demo video) are included.
- ☐ The layout is clean, with consistent typography and color palette.
- ☐ No unimplemented features (IoT, VR, AR) are implied.
- ☐ Content is concise and avoids unnecessary fluff.

8. Next tasks

After reviewing this benchmark, the recommended next agent task is:

G1_salary_from_live_job_postings

However, since the `web` tool is disabled, live job postings cannot be fetched at this time. The focus should return to tasks that can be performed locally or using pre-provided data. Consider generating or refining internal assets, such as:

- creating the technical breakdown for the Blender portrait;
- drafting the skeleton of the GitHub README for TwinSight and ARA;
- preparing the CSV for the application tracker with sample data.

This completes the **ArtStation breakdown benchmark** based on general industry patterns and best practices.

¹ Digital Twin of Production Facility — VR Training Case Study | HapzXR
<https://hapzxr.com/case-studies/digital-twin-of-production-facility/>